

EXP 14: 5G Throughput vs SNR with Modulation Labels Date

Objective 1 Matlab Code and Output

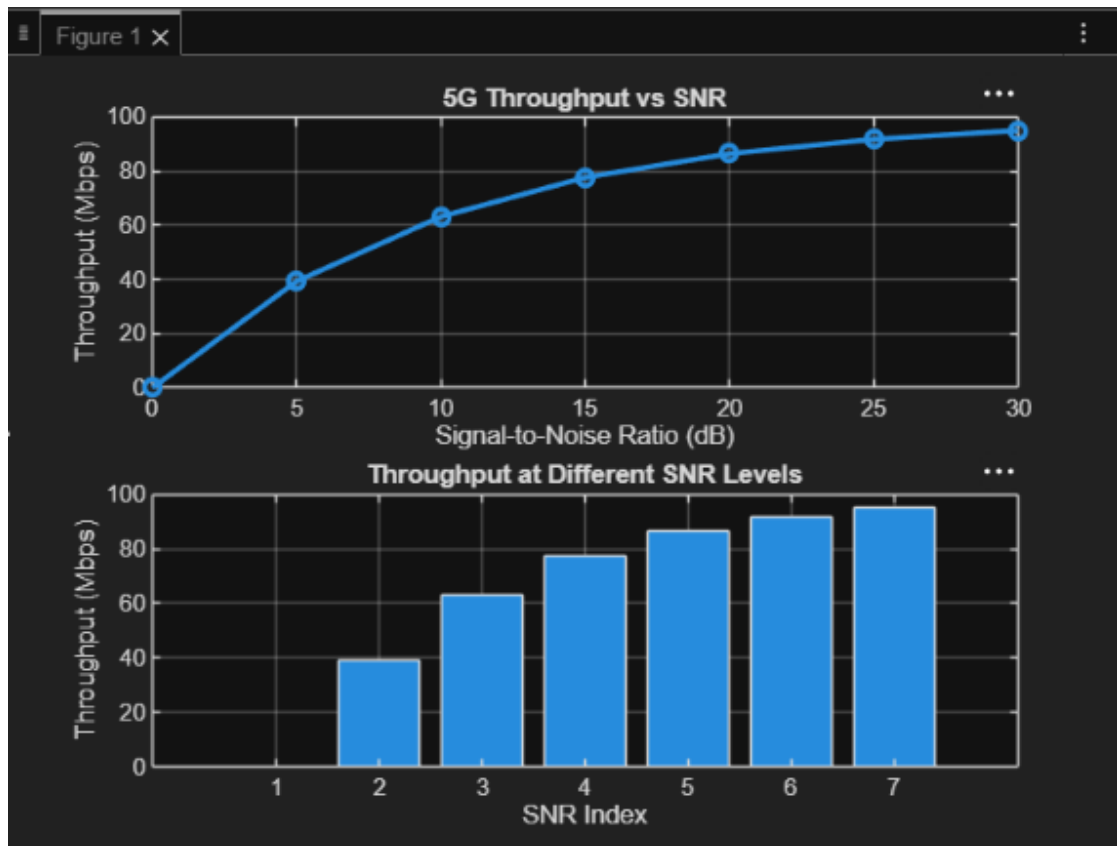
Program code:

```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Maximum Throughput (Mbps)
Max_TP = 100;
% Throughput calculation
Throughput = Max_TP*(1-exp(-SNR/10));
disp('SNR(dB) Throughput(Mbps)')
disp([SNR' Throughput'])
figure
% ----- Graph 1 -----
subplot(2,1,1)
plot(SNR,Throughput,'-o','LineWidth',2)
title('5G Throughput vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Throughput (Mbps)')
grid on
% ----- Graph 2 -----

subplot(2,1,2)
bar(Throughput)
title('Throughput at Different SNR Levels')
xlabel('SNR Index')
ylabel('Throughput (Mbps)')
grid on
```

Output:





Objective 2 Matlab Code and Output

Program code:

```

clc;
clear;
close all;

% SNR values (dB)
SNR = 0:5:30;

% Simulated Bit Error Rate (BER)
BER = [0.30 0.18 0.10 0.05 0.02 0.008 0.001];

% Reliability (%)
Reliability = (1 - BER) * 100;

disp('SNR(dB) BER Reliability(%)')
disp([SNR' BER' Reliability'])

figure
% ----- Graph 1 -----
subplot(2,1,1)
semilogy(SNR, BER, '-o', 'LineWidth', 2)
title('SNR vs Bit Error Rate in 5G')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Bit Error Rate (BER)')
grid on

```

```

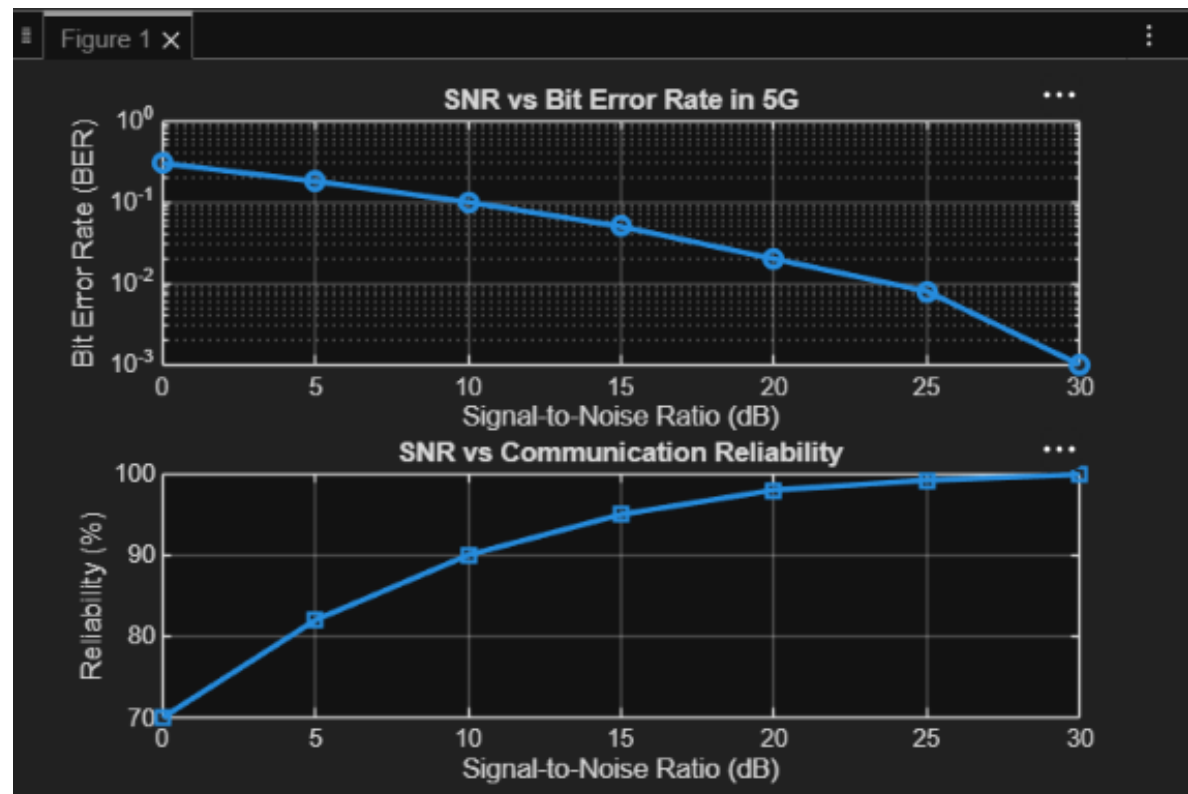
% ----- Graph 2 -----
subplot(2,1,2)
plot(SNR, Reliability, '-s', 'LineWidth', 2)
title('SNR vs Communication Reliability')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Reliability (%)')
grid on

```

Output:

Command Window			
SNR(dB)	BER	Reliability(%)	
0	0.3000	70.0000	
5.0000	0.1800	82.0000	
10.0000	0.1000	90.0000	
15.0000	0.0500	95.0000	
20.0000	0.0200	98.0000	
25.0000	0.0080	99.2000	
30.0000	0.0010	99.9000	

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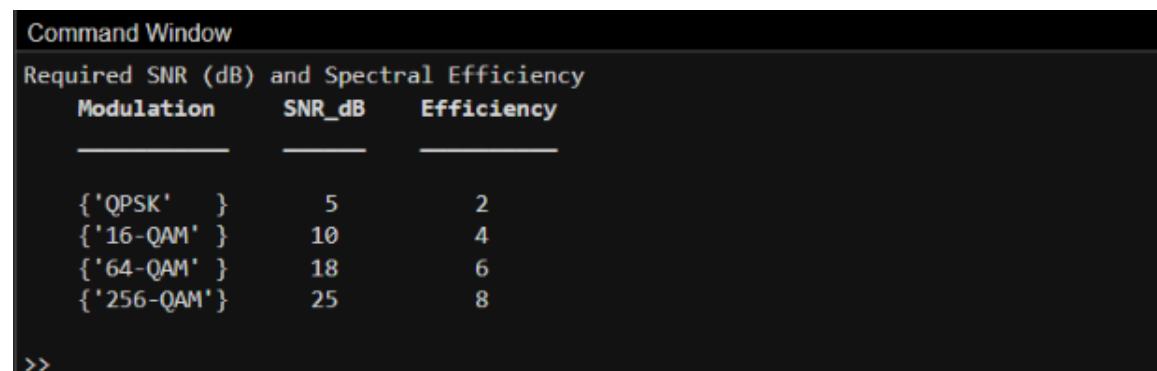
Objective 3 Matlab Code and Output

Program code:

```
clc;
clear;
close all;

% Modulation schemes
Modulation = {'QPSK', '16-QAM', '64-QAM', '256-QAM'};
% Required SNR (dB)
SNR = [5 10 18 25];
% Spectral Efficiency (bits/s/Hz)
Efficiency = [2 4 6 8];
disp('Required SNR (dB) and Spectral Efficiency')
disp(table(Modulation',SNR',Efficiency', ...
'VariableNames',{'Modulation','SNR_dB','Efficiency'}))
figure
% ----- Graph 1 -----
subplot(2,1,2)
plot(SNR,Efficiency,'-o','LineWidth',2)
title('SNR vs Spectral Efficiency')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Spectral Efficiency (bits/s/Hz)')
grid on
% ----- Graph 2 -----
subplot(2,1,1)
plot(1:4,SNR,'-s','LineWidth',2)
xticks(1:4)
xticklabels(Modulation)
title('Required SNR for Different Modulation Schemes')
xlabel('Modulation Scheme')
ylabel('Required SNR (dB)')
grid on
```

Output:



Required SNR (dB) and Spectral Efficiency		
Modulation	SNR_dB	Efficiency
{ 'QPSK' }	5	2
{ '16-QAM' }	10	4
{ '64-QAM' }	18	6
{ '256-QAM' }	25	8

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